MIL-P-85121(AS)
8 January 1980
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

PROJECTILES, 20 MM

M ARK 100 SERIES

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the characteristics of inert metal parts and their performance and test requirements for the projectiles *used* in the MARK 100 Series aircraft gun ammunition.

1.2 Classification. Projectiles covered by this specification shall be of the following types:

(a) Pressure test
(b) MARK 11 MOD 1
(c) MARK 12 MOD 0
(d) MARK 13 MOD 0
(e) MARK 14 MOD 0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1305

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2. APPLICABLE DOCUMENTS.

2.1 issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-13812	Steel Plate, Wrought, Homogeneous for Ammunition Testing (1/4 to 12 inches).
MIL-I-45607	Inspection Equipment, Supply and Maintenance, for Ordnance.
MIL-C-85122	Cartridges, 20 MM MARK 100 Series.

STANDARDS

FEDERAL

FED-STD-141	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling end Testing.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspections by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-1167	Ammunition Data Card.
MIL-STD-1168	Ammunition Lot Numbering.
MIL-STD-1695	Environments, Working, Minimum Standards for.

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DRAWINGS

Naval Sea Systems Command
(Code Ident 10001)

NAPEC-0050	20 MM MK 14, MK 13, MK 12, MK 11 "NDT" Eddy Current: Profile Cam and Standards.
NAPEC-0182	Information - For Non-Destructive Testing (NDT) of AP Projectiles.

Department of the Army
Ordnance Corps
(Code Ident 19021)

B7259545	Band Testing Machine.
D11075203	Eddy Current Flaw Detector.
7259136-5	Shipping Container for Inert Projectile, 20 MM, Assemblies and Metal Parts Assemblies (10,000).

Naval Air Systems Command
(Code Ident 30003)

DL 982432	Body Assembly, Projectile, 20 MM MARK 12 MOD 0.
DL 1350781	Body Assembly, Projectile, 20 MM MARK 13 MOD 0.
DL 1381197	Projectile, 20 MM, MARK 14 MOD 0
DL 2176062	Projectile, 20 MM, M ARK 11 MOD 1.
DL 2605727	Projectile, 20 MM, Pressure Test.

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PUBLICATIONS

MILITARY

Naval Sea Systems Corn mand
(Code Ident 10001)

NAPEC-01-2	Operation Instruction for Eddy Current Flaw Detector.

Naval Air Systems Command
(Code Ident 30003)

AD-1324	Range Firing Procedures for MARK 100 Series 20 Millimeter Naval Aircraft Gun Ammunition.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 General. The requirement for the projectile shall be as specified herein for the type projectile ordered. The projectile shall pass all applicable ballistic tests of MIL-C-85122, as specified by the procuring activity. See 6.2.1 for type of projectile ordered, applicable tests of **MIL-C-85122,** and testing activity.

3.2 First article. Unless otherwise specified in the contract or purchase order, the contractor shall furnish 500 sample projectiles for first article inspection and approval (see 4.4 and 6.2.1). The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. The samples shall be delivered to the testing activity designated in the contract or purchase order (see 6.2.1). Any production prior to acceptance of the first article sample shall be at the risk of the contractor.

3.3 Conformance to documents. The projectiles shall be manufactured in accordance with the requirements specified herein and the requirements of the documents listed on the following drawing lists (DLs):

(a) Pressure Test . . . . . DL 2605727
(b) MARK 11 MOD 1. . . . . DL 2176062

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(c) MARK 12 MOD 0 . . . . . DL 982432

(d) MARK 13 MOD 0. . . . . DL 1350781

(e) MARK 14 MOD 0. . . . . DL 1381197

3.4 Materials.

3.4.1 Chemical compositions. The projectile assembly and components shall meet the requirements of the material specification listed on the applicable drawing.

3.5 Mechanical properties.

3.5.1 Residual stress. Components produced by cold forming operations shall be adequately stress relieved to assure freedom from susceptibility to cracking or subsequent distortion or malfunction due to unrelieved stress.

3.5.1.1 Hardness. The projectile assembly and components shall meet the hardness requirements specified on the applicable drawings. M ARK 13 MOD 0 and M ARK 14 MOD 0 projectile assemblies shall also meet the penetration requirements of 3.6.1, when loaded and fired as specified in AD-1324.

3.5.2 Freedom from defects.

3.5.2.1 Soundness. Metal discontinuity in the MARK 11 MOD 1 and M ARK 12 MOD 0 projectile bodies, and the MARK 13 MOD 0 adapter, and the M ARK 14 MOD 0 windshield as determined by eddy current inspection, shall be less than that represented by production check standards of NAP EC-0050 drawings.

3.5.2.2 Fractures or inclusions. Metal discontinuity in the MARK 13 MOD 0 and M ARK 14 MOD 0 projectile bodies, as determined by ultrasonic inspection, shall be less than that represented by production check and calibration standards of NAPEC-0182 drawings.

3.5.3 Static forces. Projectile body component assemblies shall withstand the following static forces:

(a) Pressure test and M ARK 11 MOD 1 projectiles shall withstand 40 inch-pounds torque with no angular displacement (see 6.3.1) of the nose plug relative to the body.

(b) M ARK 13 MOD 0 projectile body assemblies shall withstand a compressive axial load of 8,000 pounds without a permanent increase greater than 0.0015-inch diameter in the cylindrical portion of the

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adapter. After having successfully passed the axial load test, the test samples shall withstand 2,500 inch-pounds torque with no angular displacement of the adapter relative to the body.

(c) MARK 14 MOD 0 projectiles shall withstand a compressive axial load of 6,000 pounds without a permanent increase greater than 0.001 5-inch diameter in the cylindrical portion of the windshield. After having successfully passed the axial load test, the test samples shall withstand 1,000 inch-pounds torque with no angular displacement of the windshield relative to the body.

3.5.4 Band tightness. The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the serrated band seat after knurling. The rotating band blank shall fill the band seat with a maximum allowable void of 0.004 inch between the band seat and band seat knurling.

3.6 Ballistic performance.

3.6.1 Penetration. The armor piercing tracer (APT) and armor piercing incendiary (API) project iles shall penetrate homogeneous armor plate comform ing to MIL-S-13812 under the conditions of Table 1.

TABLE 1. Penetration conditions.

Type projectile	Plate thickness (inches)	Obliquity (degrees to normal)	Striking velocity 1/ (ft/sec)
MARK 13 MOD 0 API	0.50 0.75	30 30	2,075 ± 25 2,850 ± 25
MARK 14 MOD 0 APT	0.50	30	2,600 ± 25

1/ See AD-1324 for ballistic test procedures.

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3.6.2 Accuracy. The projectiles shall meet the applicable accuracy requirements of MIL-C-85122.

3.6.3 Metal parts security. The attachment of each component to the projectile body shall be such that each component shall be capable of withstanding the firing forces experienced in service weapons without separation of any of the components.

3.7 Environmental.

3.7.1 Salt spray resistance. Pressure test and MARK 11 MOD 1 painted projectiles shall withstand 72 hours of salt spray ; and the MARK 12 MOD 0, MARK 13 MOD O, and MARK 14 MOD O painted projectiles shall withstand 144 hours of salt spray without evidence of corrosion, creepage, blistering, or loss of adhesion of the projectile coating.

3.8 Identification and marking. Projectiles shall be identified, marked, and color coded as specified on the applicable drawing.

3.9 Government-loaned property. The following equipment shall be furnished by the Government, when required (see 6.2.1).

(a) Eddy Current Flaw Detector . . . Drawing D11075203.

(b) Band Testing Machine . . . . . Drawing B7259545.

3.10 Workmanship. Projectiles shall be assembled utilizing good workmanship practices common to the ordnance industry. Particular care shall be taken that all components are clean, free from grease, dirt, rust, chips, or other foreign material. Components with obvious defects or handling damage, that would affect performance of the projectile or test weapon, shall not be assembled.

3.10.1 Special working conditions. Facilities used in the manufacture, testing, and inspect ion of projectiles shall meet the minimum standard for working conditions of MIL-STD-1695.

3.11 Ammunition data cards. When specified in the contract or purchase order, ammunition data cards (DD Form 1650) shall be completed in accordance with MIL-STD-1167 for each lot of projectiles (see 6.2.2). Distribution of ammunition data cards shall be as directed by the procuring activity (see 6.2.1).

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4. QUALITY ASSURANC E PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performancc of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Special working environment. To meet the requirements of 3.10.1, the contractor shall provide adequate facilities for the fabrication, assembly, and testing of supplies to be delivered in accordance with this specification.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) First article inspections (see 4.4).

(b) Quality conformance inspections (see 4.5).

(1) Individual inspections.

(2) Sampling inspections.

4.3 Inspection provisions, and conditions. All firing inspections (tests) shall be accomplished at the testing activity specified in the contract or purchase order (see 6.2.1). The test projectiles shall be loaded by the testing activity into the applicable MARK 100 Series cartridge configuration for ballistic tests in accordance with AD-1324.

4.3.1 inspection equipment. Unless otherwise specified in the contract or purchase order (see 6.2. 1), all inspection equipment shall be supplied and maintained by the contractor in accordance with MIL-I-45607.

4.4 First article inspection. The 500-projectile first article sample (see 3.2) shall be subjected to the examinations and tests of Table II in the sequence shown. Group A examinations and tests shall be accomplished at the contractor's facility. Group B tests shall be accomplished at a testing activity designated in the contract or purchase order (see 4.3). Failure of any projectile to meet the requirements specified herein shall be cause for rejection of the lot.

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TABLE II. First article inspection.

Test or examination	Sample	Requirement paragraph	Method paragraph
Group A			
Visual and mechanical	All	3.3, 3.8, 3.10	4.6.1
Certification	All	3.4.1, 3.5.1	4.6.1.1
Ultrasonic 1/	All	3.5.2.2	4.6.1.3
Eddy current 2/	All	3.5.2.1	4.6.1.2
Band tightness	All	3.5.4	4.6.2.1
Static forces 3/	10	3.5.3	4.6.2
Salt spray resistance	50	3.7.1	4.6.5.1
Group B			
Metal parts security	350	3.6.3	4.6.3
Penetration 1/	50	3.6.1	4.6.4
Accuracy 2/	20	3.6.2	MIL-C-85122

1/ MARK 13 MOD 0 and MARK 14 MOD 0 only.

2/ MARK 11 MOD l, MARK 12 MOD 0, MARK 13 MOD 0, and MARK 14 MOD 0.

3/ Pressure test, MARK 11 MOD 1, MARK 13 MOD 0, and MARK 14 MOD 0, as applicable.

4.5 Quality conformance inspections. The quality conformance sample shall be subjected to the examinations and tests of Table 111 in the sequence shown. Groups A and B tests shall be performed at the contractor's facility. Group C ballistic tests shall be accomplished at the testing activity designated in the contract or purchase order (see 6.2.1).

4.5.1 Production lot. A lot shall consist of projectiles produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision. The production lot, as defined by MIL-STD-1 168, shall be not less than 20,000 projectiles and not more than 50,000 projectiles.

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TABLE III. Quality conformance inspection.

Test or examination	Sample	Requirement paragraph	Method paragraph
Group A			
Visual and mechanical	All	3.3, 3.8, 3.10	4.6.1
Certification	All	3.4.1, 3.5.1	4.6.1.1
Ultrasonic 1/	All	3.5.2.2	4.6.1.3
Eddy current 2/	All	3.5.2.1	4.6.1.2
Group B			
Static forces 3/	S-2	3.5.3	4.6.2
Salt spray resistance	S-2	3.7.1	4.6.5.1
Band tightness	315	3.5.4	4.6.2.1
Group C			
Metal parts security	100	3.6.3	4.6.3
Penetration 1/	15	3.6.1	4.6.4
Accuracy 2/	20	3.6.2	MIL-C-85122

1/ MARK 13 MOD O and MARK 14 MOD O only.

2/ MARK 11 MOD 1, MARK 12 MOD 0, MARK 13 MOD 0, and MARK 14 MOD O, as applicable.

3/ Pressure test, MARK 11 MOD 1, MARK 13 MOD O, and MARK 14 MOD O, as applicable.

4.5.1.1 Lot identification. Each projectile and each packed projectile lot shall be identified in accordance with applicable drawings and MIL-STD-1168, supplemented as directed by the procuring activity (see 6.2.1).

4.5.2 Individual inspections. The individual inspections of Table III, Group A, shall be performed on the entire production lot prior to the sampling tests. Failure to meet the specific requirements shall be cause for rejection of any projectile from the lot.

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4.5.3 Lot sampling. Upon completion of the individual inspections, sample projectiles shall be randomly selected in accordance with MIL-STD-105, Inspection Level S-2, for an Acceptable Quality Level (AQL) of 1.0 percent defective and subjected to the Group B static forces and salt spray resistance tests of Table III. Three hundred fifteen projectiles shall be randomly selected from the production lot for the band tightness tests of Group B. One hundred fifty projectiles shall be randomly selected and sent to the testing activity designated in the contract or purchase order for Group C ballistic firing tests (see 6.2.1). Failure of the sample to meet the requirements of the Group C firing tests shall be cause for lot rejection.

4.6 Inspection methods.

4.6.1 Visual and mechanical. The projectile shall be visually and mechanically inspected for conformance to the applicable drawing (see 3.3), identification and marking (see 3.8), and workmanship (see 3.10).

4.6.1.1 Certification. The contractor shall have available for the Government representative written certification, accompanied by objective quality evidence (see MIL-STD-109), that the components and processes used as a part of the projectile meets the requirements specified herein. A certified analysis of each melt of metal used shall also be available to the procuring activity.

4.6.1.2 Soundness. The MARK 11 MOD 1 and MARK 12 MOD 0 bodies, MARK 13 MOD 0 adapter, and the MARK 14 MOD 0 windshield shall be inspected automatically for metal discontinuities with the eddy current flaw detector (Drawing D11075203) in accordance with NAPEC-OI-2 instructions. Projectiles failing to meet the requirements of 3.5.2.1 shall be rejected.

4.6.1.3 Fractures or inclusions. The MARK 13 MOD 0 body (Drawing 1350781, Item 1A) and the MARK 14 MOD 0 body (Drawing 1381197, Item 1A) shall be ultrasonically tested against the production standards outlined in NAPEC-0182. Projectile bodies failing to meet the requirements of 3.5.2.2 shall be rejected.

4.6.2 Static forces.

(a) Torque (pressure test and MARK 11 MOD 1). Sample projectiles shall be clamped in a suitable fixture, which can subject the contacting area of the body to the torque load. With the nose plug immobilized, apply a minimum of 40 inch-pounds torque to the projectile body. (The nose plug of the test projectiles may be modified to facilitate application of test loads.) The projectiles shall meet the requirements of 3.5.3(a).

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(b) Torque and axial load (MARK 13 MOD 0). The sample body assemblies {Drawing 1350781) shall be tested as follows:

(1) Subject the body assembly to a compressive axial load at a minimum rate of 0.25 ± 0.05 inch per minute to 8,000 pounds minimum. (Crushing of the nose end of the adapter shall not be cause for rejection.)

(2) Clamp the body assembly in a suitable fixture and subject the contacting areas of the body and adapter to 2,500 inch-pounds torque. (The adapter cavity may be plugged in order to prevent distortion of the adapter when clamped in the fixture.) (See 6.3.1.)

The body assembly shall meet the requirements of 3.5.3(b).

(c) Torque and axial load (MARK 14 MOD 0). The sample projectile assemblies (Drawing 1381197) shall be tested as follows:

(1) Subject the projectile to a compressive axial load at a rate of 0.25 ± 0.05 inch per minute to 6,000 pounds minimum. (Crushing of the windshield shall not be cause for reject ion.)

(2) Clamp the projectile in a suitable fixture and subject the contacting area of the body and the windshield to 1,000 inch-pounds torque. (The nose end of the windshield on test projectiles may be modified to facilitate application of test loads.) (See 6.3.1.)

The projectiles shall meet the requirements of 3.5.3(c).

4.6.2.1 Band tightness. Sample projectiles shall be tested using Government-furnished equipment Drawing B7259545). The sample shall be cycled through the tester to determine compliance with 3.5.4. The total measurement of radial void (band looseness) registers on the test indicator at final ram pressure, and shall be divided by four (number of indenters) to determine the average radial void. If two or more units of the sample have an average radial void between the seated band and the band seat greater than 0.004 inch, the lot shall be rejected. If one unit of the sample has an average radial void greater than 0.004 inch, a second sample consisting of the same number of units as specified for the first sample shall be tested. If, in the accumulated samples, two or more units have an average radial void greater than 0.004 inch, the lot shall be rejected.

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4.6.3 Metal parts security. The test method shall be in accordance with AD-1324. Test cartridges are to be loaded with the type end quantity of propellant, as determined by the testing activity, to provide a chamber pressure of 68,000 ± 2,000 pounds per square inch (psi) copper at 70 degrees Fahrenheit (°F) or 81,600 ± 2,500 psi, as measured by piezoelectric transducers. Half the test cartridges shall be fired at a minimum rate of 925 shots per minute (spin) from the MARK 12 Gun and the other half at a minimum rate of 3,800 spm from the MARK 4 Gun Pod. The occurrence of one or more of the following casualties, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s) shall cause rejection of the lot:

(a) Body breakup.

(b) Loss of nose, fuze, or windshield.

(c) Complete rotating band separation (see 6.3.2).

(d) Partial rotating band separation (see 6.3.3).

(e) Thrown band fragments having no evidence of band seat knurling imprint. The following criteria shall apply:

(1) The lot will be rejected if three or more assemblies throw band fragments having lengths no greater than 0.500 inch.

(2) The lot will be rejected if any assembly throws a band fragment having a length greater than 0.500 inch (indicative of deep-seated band lamination).

The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering. The cartridges shall meet the requirements of 3.6.3.

4.6.4 Penetration. API or APT projectiles shall be single fired against armor plate. The armor plate must be supported in such a manner that no vertical or horizontal span shall exceed 36 inches. The first article test shall consist of at least 20 satisfactory impacts for each test condition of 3.6.1. Thereafter, the quality conformance test shall consist of at least 10 satisfactory impacts for one test condition of 3.6.1. Complete penetration (see 6.3.4) of the armor plate by at least 70 percent of the impacts shall comprise an acceptable test. The projectiles shall meet the requirements of 3.6.1. Impacts occurring under the following conditions shall be considered unsatisfactory:

(a) Any projectile which strikes within 1 caliber of the nearest edge of the previous impact. The measurement shall be in terms of the caliber of the larger hole.

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(b) Any project ile which strikes in the spalled area of a previous impact.

(c) Any projectile which strikes within 3 calibers of the edge of the plate.

(d) Any projectile which fails to penetrate completely at the prescribed or less than the prescribed velocity. One which does satisfactorily penetrate the plate at less than the prescribed velocity shall, however, be counted as satisfactory.

(e) Any projectile which fails to impact on the plate at the prescribed angle of obliquity.

4.6.5 Environmental.

4.6.5.1 Salt spray resistance. Sample projectiles shall be tested in accordance with FED-STD-141, Method 6061 with a 5 percent salt solution for 72 hours for the pressure test and MARK 11 MOD l projectiles: and 144 hours for the MARK 12 MOD 0, MARK 13 MOD 0, and the MARK 14 MOD 0 projectiles. The projectiles shall meet the requirements of 3.7.1.

4.7 Preparation for delivery. Prior to shipment, the packaging, packing, and marking shall reinspected to ensure compliance with the requirements of Section 5.

5. PACKAGING.

5.1 Preservation, packaging, and packing.

5.1.1 Level A. Projectiles shall be packed, boxed, or palletized in accordance with Drawing 7259136-5 or as specified in the contract or purchase order (see 6.2.1).

5.2 Marking. Projectile containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The projectiles described herein are intended for use in the 20 MM MARK 100 Series aircraft gun ammunition.

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6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of projectile ordered (see 3.1).

(c) Ballistic tests required (see 3.1).

(d) Activity designated for ballistic tests (see 3.1, 3.2, 4.3, 4.5, and 4.5.3).

(e) When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Aquisition Regulations (DAR). The first article should be a preproduction sample consisting of 500 projectiles as specified in 3.2. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests and approval of the first article.

(f) Government-loaned property (see 3.9 and 4.3.1).

(g) Distribution of ammunition data cards (DD Form 1650) (see 3.11)=

(h) Identification of projectiles and projectile lots (see 4.5.1.1).

(i) Packing requirements (see 5.1).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

Paragraph	Data Requirement	Applicable DID
3.11	Ammunition Data Card	DI-L-1410

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.3 Definitions.

6.3.1 Angular displacement. Angular displacement is determined by means of measurement of a prescribed mark on the body and the nose of the projectile (parallel to the longitudinal axis). Mismatch of the scribed line by over 0.010 inch after the test is defined as angular displacement.

6.3.2 Complete rotating band separation. Detachment of all of the band having band seat knurling imprint.

6.3.3 Partial rotating band separation. Detachment of part of the band having band seat knurling imprint.

6.3.4 Penetration. The entire projectile or major portion of the projectile passes completely through the armor plate.

6.4 Supersession data. This specification includes the requirements of purchase description AS-2703A dated 12 August 1976.

Preparing activity:
NAVY - AS

Project number: 1305 -N834

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-85121(AS) Projectiles, 20 MM MARK 100 Series

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

[ ] VENDOR [ ] USER [ ] MANUFACTURER

1 [ ] HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? [ ] IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

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